

JOHN LEWIS

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EDUCATION

University of Virginia, Charlottesville, VA

August 2024 - May 2028

B.S. Mechanical Engineering, Minor in Engineering Business

- Activities: UVA Motorsports, UVA Gizmologists, Virginia Area Ski and Snowboard Team, UVA Men's Rugby
- Relevant Coursework: Thermodynamics, Applied Circuits, Strengths of Materials, Dynamics, Mechanics Lab, Computer-Aided Design

Albemarle High School, Charlottesville, VA

August 2020 - May 2024

RELEVANT EXPERIENCE

Team Lead, Potential Energy Propulsion Racer, Charlottesville, VA

October 2025 - December 2025

- Designed and modeled a spring and gravity-powered vehicle in SolidWorks, developing all parts and assemblies under strict material and size constraints
- Iteratively designed and refined a spur-gear gearbox, landing on a 12:1 gear ratio to maximize mechanical advantage and efficiently convert spring potential energy into wheel torque
- Analyzed gear ratio trade-offs between torque, speed, and traction, optimizing drivetrain performance for obstacle completion and jump distance

Design Lead, Pine Xylem as a Method of Removing E. coli from Unclean Water, Charlottesville, VA

January 2025 - May 2025

- Developed a natural method of filtering unclean water, capable of removing 99% of E. coli bacteria in water
- Collaborated with a team of 5 to implement use of a Pine Xylem puck along with pumice and charcoal to devise a natural filter
- Made significant use of Fusion 360, additive manufacturing, and practiced effective team communication

Mechanical Lead, Gizmologists: University of Virginia, Charlottesville, VA

August 2024 - Present

- Led a team of 15+ members in mechanical design decision-making, serving as the primary authority on MechE systems and CAD organization
- Designed and built a clock with a gantry, stepper motors, and servo motor with a stylus to write time on pressure-sensitive boards using an Arduino microcontroller and C/C++
- Designed and constructed simple button game to bring to local STEM fairs and schools in the Charlottesville area to demonstrate STEM skills while incorporating fun
- Developed and communicated club objectives, such as training new members to improve project participation and progressing on design projects

Member: Chassis, Virginia Motorsports SAE Team, Charlottesville, VA

August 2024 - Present

- Researched and modeled optimal chassis design within constraints set by competition.
- Performed multiple FEA simulations to ensure chassis will not fail, and keep driver safe
- Gained experience programming and using CNC Mills, welding, and operating a Water Jet to manufacture parts for the chassis

LEADERSHIP EXPERIENCE

Mechanical Lead, AHS FIRST Robotics Club, Charlottesville, VA

September 2021 - June 2024

- Directed sub-team of 6 people to design and construct chassis and mechanical parts of robot, following specific constraints
- Held courses in Fusion 360 to inexperienced team members
- Led multiple demonstrations and held STEM workshops in local elementary schools to promote STEM

Varsity Team Captain, Albemarle Rowing, Charlottesville, VA

August 2020 - June 2024

- Commit 15+ hours weekly to practices and competition year-round
- Created training plans for fellow teammates in off seasons, leading to 15 out of 17 boats achieving a state title

WORK EXPERIENCE

Server and Bartender, Shelter Island Yacht Club, Shelter Island, NY

June 2024 - August 2025

- Cooperated closely with a staff of 30 to ensure an efficient workplace
- Recorded food and drink orders and served customer requests to the highest standards
- Maintained full product knowledge of menu, specials, and cocktail recipes

SKILLS & CERTIFICATIONS

- Technical: SolidWorks, Fusion 360, MS Office, Autodesk CFD
- Programming Languages: MATLAB, C++, Python, Java, C#
- Certifications: Certified SolidWorks Associate obtained through *Dassault Systems* in October 2025